LIN 313: Language and Computers

Practice Review Exam, Fall 2026

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Instructions

- This practice exam is **longer** than the real midterm. Use it to find the topics you need to review.
- To practice under exam conditions, put away your devices and allow yourself one page (both sides) of notes.
- For calculation questions, **show your work**. You do not need to reduce fractions or calculate exact decimals (e.g., leaving an answer as $\frac{2}{7}$ or $\log_2(6)$ is completely acceptable).
- Please **box your final numerical answers**.

Part 1: Conceptual True / False

*Circle **T** for True or **F** for False for each statement below.*

1. [**T** / **F**] Onomatopoeic words (like *buzz* and *hiss*) and iconic signs in sign languages are exceptions to Hockett's design feature of *arbitrariness*.
2. [**T** / **F**] In repeated reference games (like the Tangrams communication experiment), descriptions produced by a pair typically become longer and more verbose as the rounds progress.
3. [**T** / **F**] ASCII uses 7 bits per character, so it can represent 128 distinct characters.
4. [**T** / **F**] A fair coin has lower entropy than a biased coin that lands heads 80% of the time.
5. [**T** / **F**] Entropy is an upper bound on compression: a clever enough code can always use fewer bits per symbol, on average, than the entropy of the source.
6. [**T** / **F**] While building a language model, we compare versions on a separate *development (dev) set*, and evaluate on the *test set* only once (or a few times) at the very end.
7. [**T** / **F**] If you evaluate an n-gram model on the same sentences it was trained on, its perplexity will look misleadingly low.
8. [**T** / **F**] As a corpus gets larger, the number of distinct word types keeps growing, so no matter how big a word vocabulary is, there will always be words it misses (like *ChatGPT* before 2023).
9. [**T** / **F**] Chinese, Japanese, and Thai mark word boundaries with spaces, so whitespace tokenization works well for them.
10. [**T** / **F**] In English, every orthographic word (a string between two spaces) corresponds to exactly one grammatical word.
11. [**T** / **F**] The Turkish word *evlerden* ("from the houses"), built from *ev* + *ler* + *den*, is an example of agglutinative morphology.
12. [**T** / **F**] Byte-Pair Encoding is designed to find morphemes: its merge rules are chosen so that each new token corresponds to a meaningful unit like a root or an affix.

Part 2: Conceptual Multiple Choice

Circle the single best answer for each question below.

1. Your friend says, "Ugh, my roommate is *so* doomscroll-y today," and you immediately understand them, even though you have never heard *doomscroll-y* before. Which of Hockett's design features does this illustrate?

   (A) Displacement
   (B) Productivity
   (C) Discreteness
   (D) Prevarication

2. Which pair of English words is a **minimal pair** showing that /p/ and /b/ are different phonemes?

   (A) *spin* and *pin*
   (B) *pat* and *pit*
   (C) *pat* and *bat*
   (D) *bat* and *bad*

3. The newspaper headline *"Squad Helps Dog Bite Victim"* has two readings. What is the main source of the ambiguity?

   (A) *bite* can be a verb (the dog bites the victim) or part of the noun phrase *dog bite victim*.
   (B) *squad* has two unrelated meanings (polysemy).
   (C) The sentence violates the *productivity* design feature.
   (D) The headline has a long-distance dependency that the reader cannot track.

4. For which blank would a good language model have the **lowest** perplexity (i.e., the smallest "effective" number of options)?

   (A) I just bought a new _____.
   (B) She said _____.
   (C) The capital of France is _____.
   (D) My favorite thing is _____.

5. As words become very common, speakers often shorten them: *laboratory* → *lab*, *mathematics* → *math*, *telephone* → *phone*. Which idea from class best explains this pattern?

   (A) Language is structured for efficient communication: frequent, predictable words carry less information each time they're used, so giving them short forms saves effort overall.
   (B) Byte-Pair Encoding merges frequent pairs of characters, which makes frequent words shorter.
   (C) Arbitrariness: there is no connection between a word's form and its meaning, so its length can change at random.
   (D) Displacement: words used to talk about things that aren't present tend to get shorter.

6. According to the chain rule of probability, which expression is **exactly** equal to $P(w_1, w_2, w_3)$, with no simplifying assumptions?

   (A) $P(w_1) \cdot P(w_2) \cdot P(w_3)$
   (B) $P(w_1) \cdot P(w_2 \mid w_1) \cdot P(w_3 \mid w_2)$
   (C) $P(w_1) \cdot P(w_2 \mid w_1) \cdot P(w_3 \mid w_1, w_2)$
   (D) $P(w_3 \mid w_1, w_2)$

7. Unicode can represent the characters of almost every writing system. Why don't language models simply use individual Unicode characters as their tokens?

   (A) Unicode cannot represent emoji.
   (B) Character sequences are very long, and the model has to learn for itself how characters combine into meaningful units.
   (C) Characters cannot be converted into binary.
   (D) There are more Unicode characters than there are words in any language.

8. The same question ("What is the capital of Morocco?") is split into many more tokens in Greek than in English by a commercial LLM's tokenizer. What is the most direct consequence for Greek speakers?

   (A) Their questions will be answered more accurately.
   (B) They will pay more and wait longer for the same amount of content.
   (C) Their text cannot be represented by the model at all.
   (D) Nothing, since the number of tokens does not matter.

Part 3: Language, Probability, and n-gram Behavior

Q3.1. Name the Hockett design feature that each example illustrates.

(a) Everyone in your English-speaking community uses *umbrella* to mean the same kind of object, and nobody uses it to mean a spoon.

Semanticity

(b) A child says "I already brushed my teeth!" when they haven't.

Prevarication

(c) The sounds at the start of *bat* and *pat* lie on an acoustic continuum, but listeners hear each one as clearly /b/ or clearly /p/, never something in between.

discreteness

Q3.2. Probability basics.

(a) A fair coin is flipped 3 times. What is $P(\text{H, H, T})$, i.e., heads, then heads, then tails?

$P(H) \cdot P(H \mid H) \cdot P(T \mid H, H)$

$\cdot\ P(H) \quad P(T) \quad \Rightarrow \frac{1}{2} * \frac{1}{2} * \frac{1}{2} = \frac{1}{8}$

(b) In a game of Wordle, 64 candidate words remain. 16 of them contain the letter s, 24 contain the letter t, and 8 contain both. Compute $P(\text{contains } s \text{ and } t)$ and $P(\text{contains } t \mid \text{contains } s)$.

$P(s \cap t) = \frac{8}{64} = \frac{1}{8}$

$P(t \mid s) = \frac{P(S \cap T)}{P(S)} = \frac{8/64}{16/64} = \frac{1}{2}$

(c) Are "contains s" and "contains t" independent in this list? Explain using your numbers.

$P(t \mid s) = P(t)$

$\frac{1}{2} = \frac{24}{64} \neq \frac{1}{2}$ → Not Ind.

Q3.3. **A smoothed n-gram model is trained on ten years of posts from the subreddit r/UTAustin. Rank these three test sets from lowest to highest perplexity, and explain your ranking in one or two sentences: (i) a 19th-century French novel, (ii) new posts from r/UTAustin, (iii) UT Austin news press releases.**

(ii) < (iii) < (i)



Q3.4. **Consider the sentence *"The keys to the old wooden cabinet _____ on the table."* To choose correctly between *is* and *are*, a model needs to see the word *keys*. What is the smallest n for which an n-gram model has *keys* in its context when predicting the blank? Why don't we just use n-gram models with a very large n?**

n = 7.

Storage explodes

Q3.5. **A bigram model is trained on just two sentences: `<s> dogs chase cats </s>` and `<s> cats chase mice </s>`. For each sentence below, say whether the model could generate it (i.e., give it a probability greater than 0), and briefly explain why.**

(a) `<s> dogs chase mice </s>`

Yes All bigrams exist

(b) `<s> cats chase cats </s>`

Yes

(c) `<s> mice chase dogs </s>`

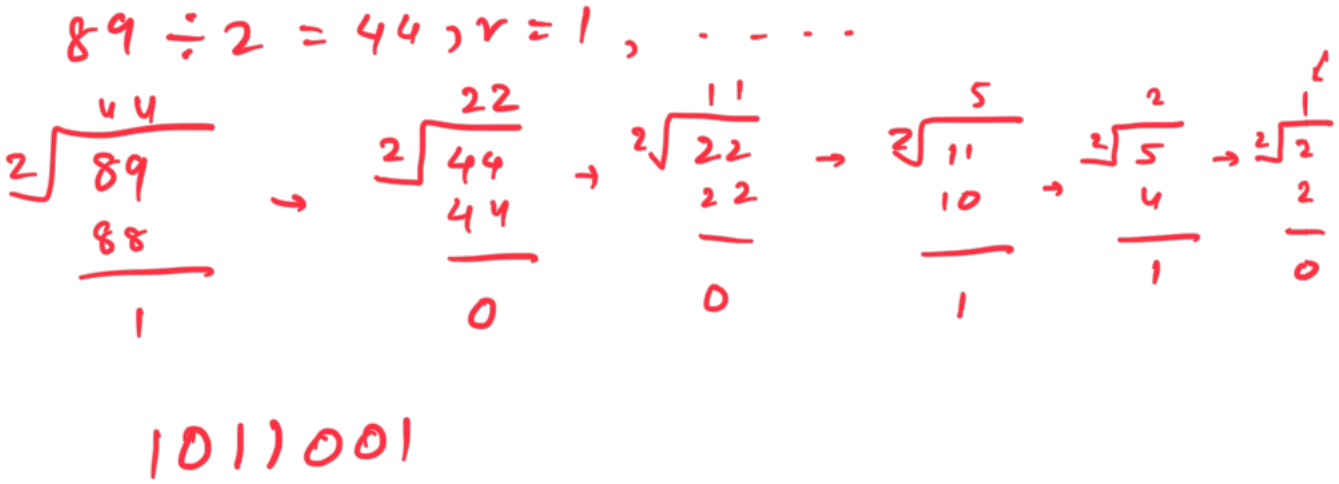
No $P(S)=0$ Prob. of Sent $= 0$. because <s> mice doesnt exist.

(d) `<s> dogs chase </s>`

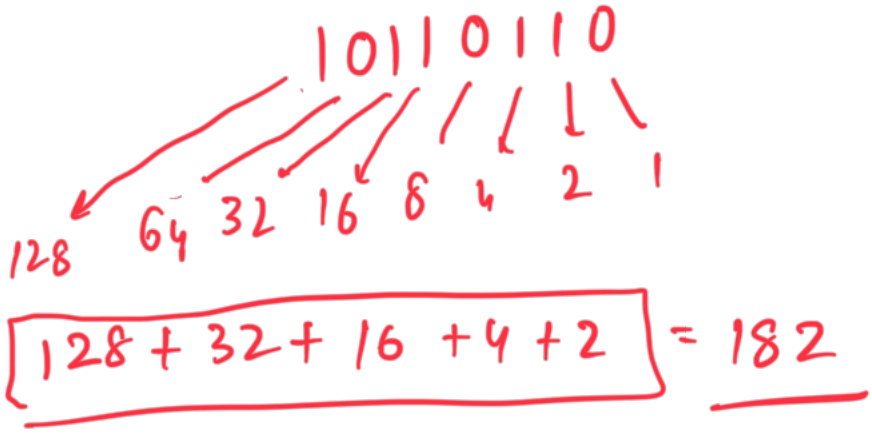
No $P(\text{</s>} \mid \text{chase}) = 0$.

Part 4: Encoding and Compression

Q4.1. Convert the decimal number 89 into binary. Show your work.



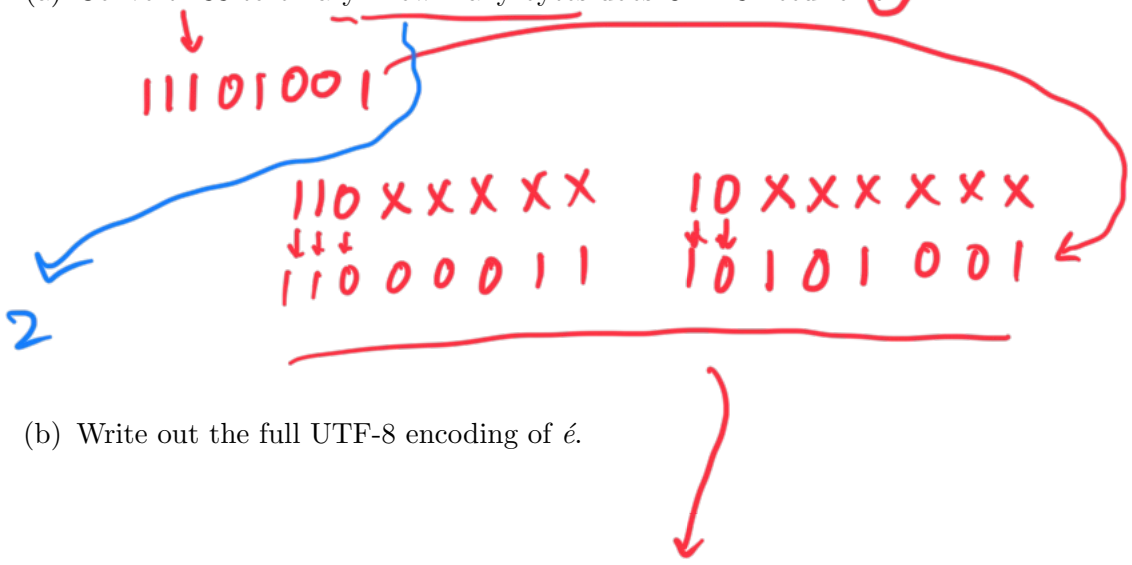
Q4.2. Convert the binary byte `10110110` to decimal. Show your work.



Q4.3. The UTF-8 templates are shown below. The character *é* has the Unicode value 233 (in decimal).

Bytes	Template	Free slots
1	`0xxxxxxx`	7
2	`110xxxxx 10xxxxxx`	11
3	`1110xxxx 10xxxxxx 10xxxxxx`	16

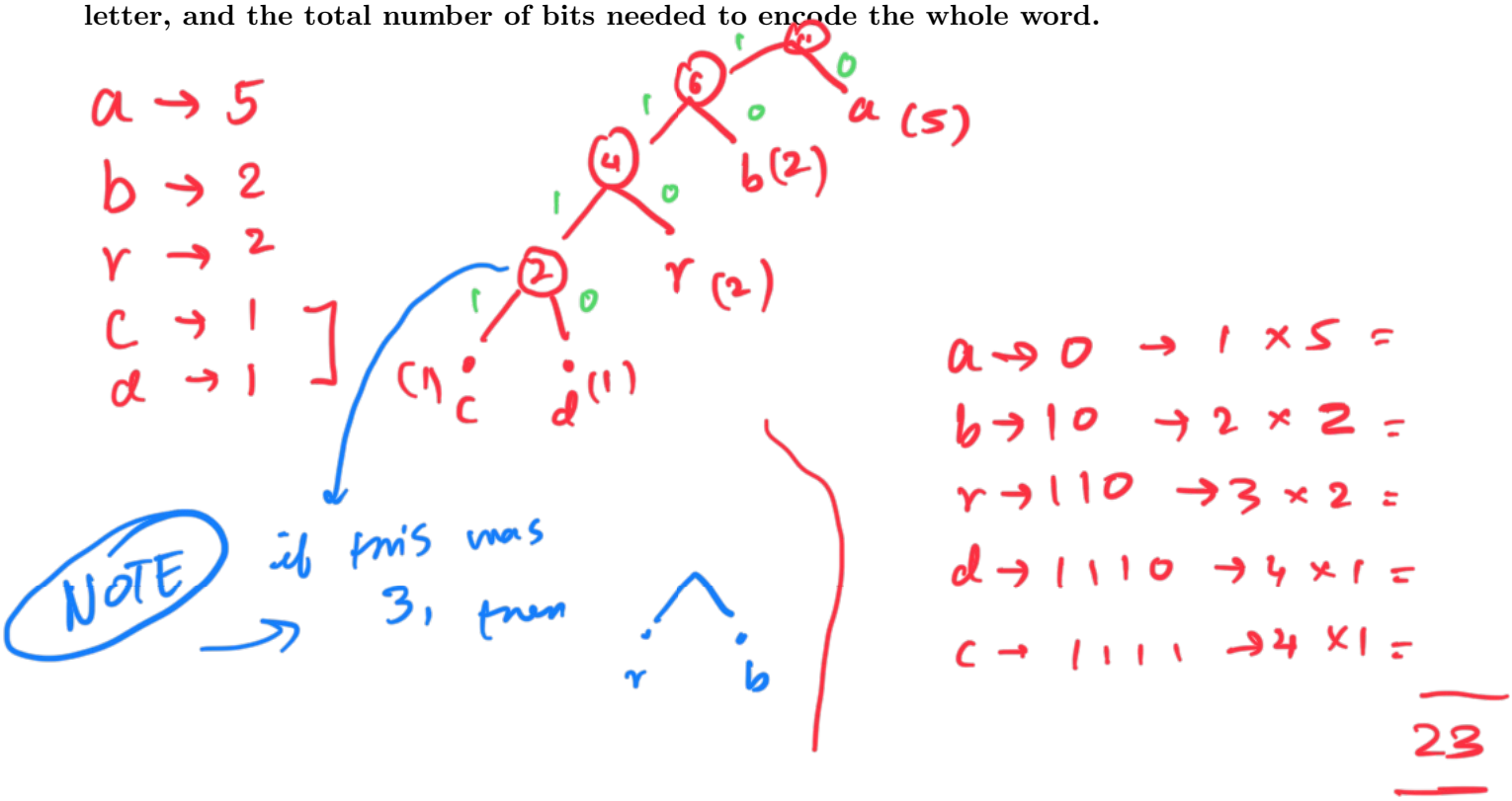
(a) Convert 233 to binary. How many bytes does UTF-8 need for *é*?



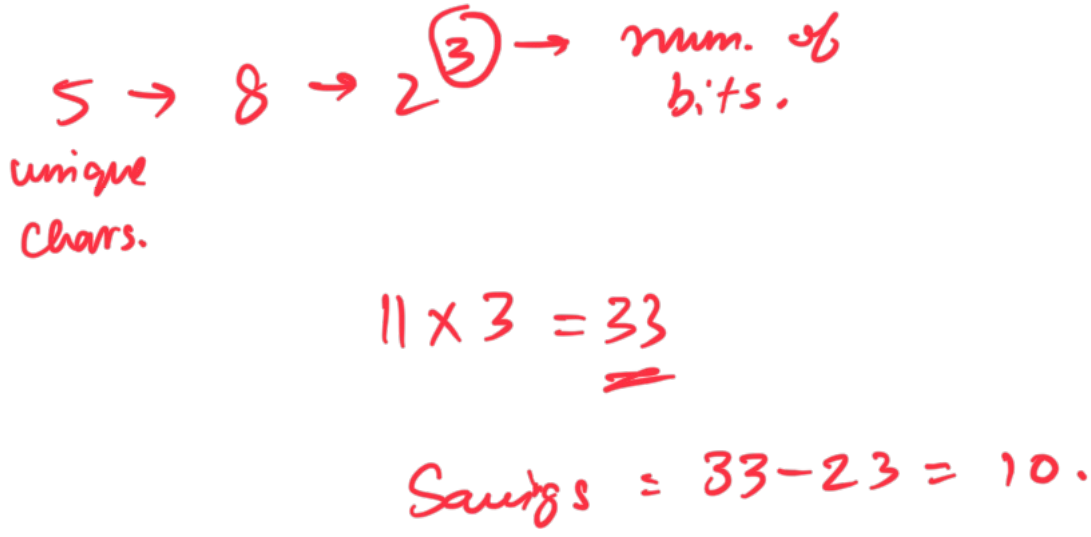
(b) Write out the full UTF-8 encoding of *é*.

(c) A decoder reads a byte that starts with `0`. What does it immediately know about that byte, and why does this make UTF-8 backwards compatible with ASCII?

Q4.4. Consider the word `abracadabra`. Compute the frequency of each letter, build a Huffman tree by hand, and assign a binary code to each letter. Report the code for each letter, and the total number of bits needed to encode the whole word.

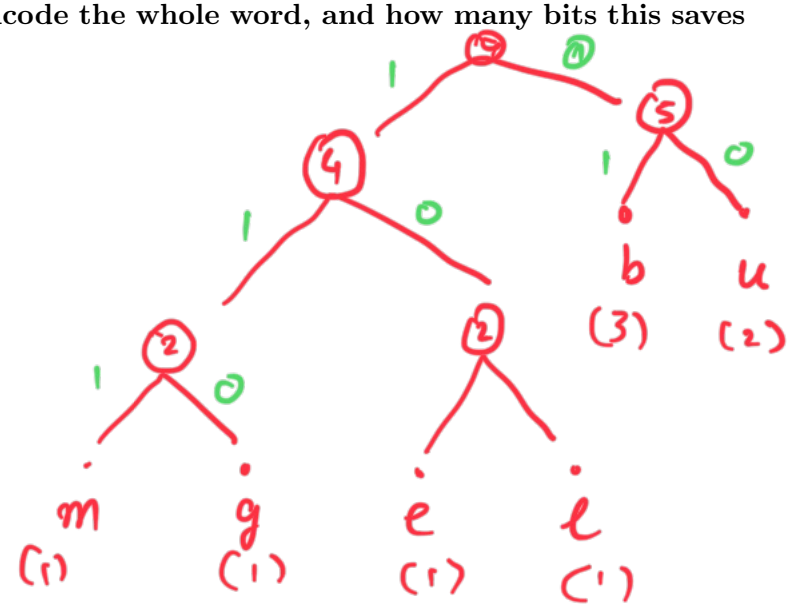


Q4.5. How many bits would a fixed-length code need to encode `abracadabra`? How many bits does the Huffman code save?



Q4.6. Now build a Huffman tree for the word `bubblegum`. Report the code for each letter, the total number of bits needed to encode the whole word, and how many bits this saves compared to a fixed-length code.

b → 3
u → 2
l → 1
e → 1
g → 1
m → 1



Q4.7. Someone proposes the code A = `0`, B = `01`, C = `10`, D = `11`. Is this code prefix-free? If it is, explain why decoding is never ambiguous. If it isn't, give a bit string that can be decoded in more than one way.

0 1 0 → A | C
0 | 1 0

or 0 1 | 0
B A

A 0
B 10
C → 1 1 0
D → 1 1 1

Part 5: Information Theory and Language

Q5.1. Austin weather. Suppose the weather on any day in Austin follows this distribution:

Weather	sunny	cloudy	rainy	stormy	hail
P	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{16}$

(a) How much information (in bits) do you get from learning that today is *sunny*? That today has *hail*?

$I(\text{sunny}) = -\log_2\left(\frac{1}{2}\right) = 1$

$I(\text{hail}) = -\log_2 \frac{1}{16} = 4$

(b) Compute the entropy of the weather distribution.

$\sum p(x) \cdot -\log_2 P(x)$

$= \frac{1}{2} \cdot 1 + \frac{1}{4} \cdot 2 - - - -$

(c) Build a Huffman code for the five weather types, and compute its average number of bits per day. How does this compare to the entropy, and to a fixed-length code?

Q5.2. Wordle, simplified. Suppose there are 64 equally likely possible answers left. You are considering two guesses. Each guess sorts the 64 candidates into groups by the color pattern you would see:

- **Guess A** produces 4 possible patterns, leaving 32, 16, 8, and 8 candidates.
- **Guess B** produces 4 possible patterns, leaving 16, 16, 16, and 16 candidates.

(a) You play Guess A and see the pattern that leaves 8 candidates. How many bits of information did you get?

$$-\log_2 \frac{8}{64}^1 = 3 \text{ bits.}$$

(b) Compute the expected information (entropy) of Guess A and of Guess B.

$$A: \frac{32}{64}^1 \cdot 1 + \frac{16}{64}^1 \cdot 2 + \frac{8}{64}^1 \cdot 3 + \frac{1}{8} \cdot 3$$

B: ~~

(do at home)

(c) Which is the better guess, and why?

B

Q5.3. Efficient sibling terms. English uses *brother* and *sister* regardless of age, but many languages (e.g., Mandarin) distinguish *older* and *younger* siblings. Suppose that speakers need to refer to each kind of sibling, older brother (OB), younger brother (YB), older sister (OS), and younger sister (YS), with these "need probabilities":

Sibling	OB	YB	OS	YS
Need probability	0.4	0.1	0.4	0.1

As in class, the *cost* of referring to sibling k is the number of bits a listener still needs to figure out which sibling was meant after hearing the word: 0 bits if the word picks out one sibling type, 1 bit if it covers two equally plausible types. The communicative cost of a system is $\sum_k P(k) \cdot \text{cost}(k)$.

(a) Compute the communicative cost of each system:
- **System 1** (English-like): *brother* = {OB, YB}, *sister* = {OS, YS}.
- **System 2**: *older brother*, *older sister*, and *younger sibling* = {YB, YS}.
- **System 3**: *older sibling* = {OB, OS}, *younger brother*, and *younger sister*.

S1: 1 bit × 0.4 + 1 × 0.1 + 1 × 0.4 + 1 × 0.1

1 bit

P(k)

S2: Cost(OB) × 0.4 + Cost(OS) × 0.4 +

0

Cost(YB) × 0.1 + Cost(YS) × 0.1 = 0.2 bits

S3: [Cost(OB) × 0.4] × 2 + [Cost(YB) × 0.1] × 2 = 0.8 bits

1 0 0 1

(b) A Mandarin-like system with four separate words has a cost of 0. Why don't all languages simply use four words (or more) for every kind of relative?

it trades off with complexity.

Q5.4. Same frequency, different information. Two words, **Word 1** and **Word 2**, each occur 16 times in a corpus, so they have exactly the same frequency. Each word occurs in two different contexts. The tables show how many times each word occurs in each context, and how predictable the word is ~~there~~ in the context.

Word 1		
Context	Count	$P(\text{Word 1} \mid \text{context})$
Context A	8	$\frac{1}{2}$
Context B	8	$\frac{1}{4}$

Word 2		
Context	Count	$P(\text{Word 2} \mid \text{context})$
Context C	8	$\frac{1}{32}$
Context D	8	$\frac{1}{64}$

(a) A word's *average information in context* is its surprisal, $-\log_2 P(w \mid \text{context})$, averaged over all of its occurrences. Compute it for Word 1 and for Word 2.

$$\frac{8 \times -\log_2 P(w_1 \mid C_A) + 8 \times -\log_2 P(w_1 \mid C_B)}{16}$$

$$\frac{8 \times 1 + 8 \times 2}{16} = \frac{8+16}{16} = 1.5 \text{ bits}$$

$$\frac{8 \times 5 + 8 \times 6}{16} = 5.5$$

(b) Piantadosi, Tily, and Gibson (2011) found that a word's average information in context predicts its length better than its frequency does. Based on your answers, which of these two words would they predict should tend to have the **longer** form, and why?

W_2 is longer

because it's less predictable

or

more informative.

Part 6: Probability and n-gram Language Models

Q6.1. Statistical learning, like a baby. An infant hears the following stream of syllables (with no pauses), which ends after the final *mo*:

`ne mo ta fi lu go ta fi ne mo lu go ne mo ta fi lu go ne mo`

For a syllable x, the transitional probability $P(y \mid x) = \frac{\text{count}(x\ y)}{\text{count}(x \text{ followed by any syllable})}$.

(a) Compute $P(mo \mid ne)$ and $P(ta \mid mo)$.

$= \frac{P(ne, mo)}{P(ne)} = \frac{4}{4} = 1$

$P(ta \mid mo) = \frac{P(mo, ta)}{P(mo)} = \frac{2}{4} = \frac{1}{2}$

(b) Based on transitional probabilities, what are the three "words" of this language?

nemo, tafi, lugo

(c) At test, the infant hears *tafi* and *mota*. Which would you expect them to listen to longer, and why?

mota because its more surprising! OR less predictable.

Consider the following 5-line corpus. Each line is padded with <s> and </s> as shown.

```
<s> i like green tea </s>
<s> i like black coffee </s>
<s> you like green tea </s>
<s> i drink green tea </s>
<s> you drink black tea </s>
```

Q6.2. Compute the following from the corpus. Show the counts you used.

(a) $P(\text{i} \mid \text{<s>}) = \frac{P(\text{<s>}, \text{i})}{P(\text{<s>})} = \frac{3}{5}$ ✓

(b) $P(\text{green} \mid \text{like}) = \frac{P(\text{like}, \text{green})}{P(\text{like})} = \frac{2}{3}$

(c) $P(\text{green} \mid \text{i like}) = \frac{P(\text{i like green})}{P(\text{i like})} = \frac{1}{2}$

if Bigram, $P(\text{green} \mid \text{like}) = 2/3$

(d) $P(\text{tea} \mid \text{like green})$

No Assumptions $= \frac{P(\text{like green tea})}{P(\text{like green})} = \frac{2}{2} = 1$

if bigram $= P(\text{tea} \mid \text{green}) = \frac{P(\text{green}, \text{tea})}{P(\text{green})}$

Q6.3. Under a bigram model, write out and compute P(<s> you like black tea </s>). $= 1$

$$\frac{P(\text{<s>}, \text{you})}{P(\text{<s>})} \times \frac{P(\text{you}, \text{like})}{P(\text{you})} \times \frac{P(\text{like black})}{P(\text{like})}$$

$$\times \frac{P(\text{black tea})}{P(\text{black})} \times \frac{P(\text{tea}, \text{<s>})}{P(\text{tea})}$$

Q6.4. Smoothing.

(a) What probability does the unsmoothed bigram model assign to <s> you like coffee </s>? Why?

0 → P(coffee | like) = 0

(b) Now apply add-one (Laplace) smoothing: $P_{\text{add-1}}(w \mid w_{\text{prev}}) = \frac{\text{count}(w_{\text{prev}}, w)+1}{\text{count}(w_{\text{prev}})+V}$, using vocabulary size $V = 9$ (the 8 word types plus </s>). Compute $P_{\text{add-1}}(\text{coffee} \mid \text{like})$ and $P_{\text{add-1}}(\text{green} \mid \text{like})$. What happened to $P(\text{green} \mid \text{like})$ compared with Q6.2(b)?

After smoothing, P(coffee | like) = 0 → P(green | like)

$= \frac{0+1}{\text{Count}(\text{like})+9} = \frac{1}{3+9} = \frac{1}{12}$

$\frac{2+1}{3+9} = \frac{3}{12} = \frac{1}{4}$

Q6.5. Perplexity. Two models predict the same 3-word test sentence. Model A assigns the three words probabilities $\frac{1}{2}$, $\frac{1}{8}$, and $\frac{1}{4}$. Model B assigns $\frac{1}{2}$, $\frac{1}{2}$, and $\frac{1}{2}$. Compute the perplexity of each model on this sentence. Which model is better?

$2^{-\frac{1}{N}\text{SUM}(\log_2 P(w))}$

$M_A \Rightarrow 2^{\frac{1}{3}(1+3+2)} = 2^{\frac{1}{3}(6)} = 2^2 = 4$

$M_B \Rightarrow 2^{\frac{1}{3}(1+1+1)} = 2^1 = 2$

Q6.6. The corpus contains *black coffee*, *black tea*, and *green tea*, but never *green coffee*. A friend says, "Since coffee and tea show up in such similar contexts, a bigram model should figure out that *green coffee* is probably fine too." In 1–2 sentences, explain why the friend is wrong about n-gram models.

→ ngram models don't learn about similarity.

OR don't track similarities and only look at freqs.

Part 7: Tokenization and Byte-Pair Encoding

Q7.1. How many word tokens (instances) and how many word types are in the sentence below? (Ignore capitalization.)

The cat saw the other cat and the dog

the other cat and saw dog

6 types 9 tokens.

Consider the following *training corpus*:

```
rain train rain brain rains train rain brain train rain rains grain
```

Use a beginning-of-word marker _ at the start of every word (including the first). Break ties alphabetically by the first symbol, then by the second, with _ coming before all letters.

Q7.2. Give the word frequency table (with each word split into symbols) and the initial vocabulary.

_, r, a, i, n 4

_, t, r, a, i, n 3

_, b, r, a, i, n 2

_, r, a, i, n, s 2

_, g, r, a, i, n 1

_, r, a, i, n, t, b, s, g

Reminder, the corpus:

```
rain train rain brain rains train rain brain train rain rains grain
```

Q7.3. Run BPE for 4 merges. For each step, report the most frequent pair with its count and the new token. Then give the final vocabulary.

① i, n → in

- , r, a, in → 4
- , t, r, a, in → 3
- , b, r, a, in → 2
- , r, a, in, s → 2
- , g, r, a, in → 1

r, a → ra

OR

② a, in → ain

- , r, ain 4
- , t, r, ain 3
- , b, r, ain 2
- , r, ain, s 2
- , g, r, ain 1

r, ain (12 times)

r, ain → "rain" ③

- , rain 4
- , t, rain 3
- , b, rain 2
- , rain, s 2
- , g, rain 1

- , rain → -rain ④

- , r, a, i, n, t, b, g, s, in, ain, rain, -rain

Q7.4. Tokenize the test corpus `strain grains ran` using your 4 merges, applied in order. Separate tokens with commas.

-, s, t, r, a, i, n
-, g, r, a, i, n, s
-, r, a, n
leave alone

1. i, n → in
2. a, in → ain
3. r, ain → rain
4. -, rain → -rain | a

⇒ -, s, t, rain, -, g, rain, s, -, r, a, n

Q7.5. What goes wrong if this tokenizer has to tokenize the word `rainy`?

-, r, a, i, n, -?? "y is not in vocab!"

Q7.6. BPE learned the token `rain` and uses it inside `train`, `brain`, and `strain`. Is `rain` a morpheme in those words? What does this tell you about the tokens BPE learns?

rain in those words is NOT a morpheme.

⇒ it is not guaranteed that BPE always learns morphemes!

FIN